

Alpilla 600 MW Combined Cycle Power Plant

Plant Heat and Mass Balances

Document number	ALP-EPC-00000-PR-HMB-0001
Description	9 cases
Revision / status	Rev A - IFR, draft (not issued)
Originator	Istanbul EPC Mühendislik ve Taahhüt A.Ş.
Discipline / type	process / HMB Heat and mass balance
Review class	approval (Basic Design Package)
Basis	requirement:ER-03.02, requirement:ER-04.06

1. Purpose and basis

This document gives the plant heat and mass balances of all design and guarantee cases: the power island performance data of Imaginary Electric, the EPC piping pressure and temperature drops, the balance-of-plant auxiliary loads and the other plant fuel consumers. Every stream enthalpy is recomputed (IAPWS-IF97 for water / steam, ideal-gas mixtures for air, fuel and flue gas), every node is closed for mass and energy, and each case is checked against the guarantees and limits it covers. Power island data:

- SRC-IECE-0005 Power island off-design heat balances (case book) - Rev A
- SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)

2. Cases

Case	Description	Fuel	Ambient degC	RH %	Seawater degC	GT load %
MAX-NG-100	Maximum ambient (40 degC) with maximum seawater (27 degC), natural gas, base load	gas	40	35	27	100
MIN-NG-100	Minimum ambient for full capability (-8 degC), natural gas, base load	gas	-8	80	6.9	100
SRC-LDO-100	Site Reference Conditions, LDO with water injection, base load (guarantee PG-15/16)	LDO	15	70	16	100
SRC-NG-060	Site Reference Conditions, natural gas, 60 % plant net load (guarantee PG-03)	gas	15	70	16	55.78
SRC-NG-100	Site Reference Conditions, natural gas, base load (guarantee case)	gas	15	70	16	100
SRC-NG-GT80	Site Reference Conditions, natural gas, GT at 80 % load	gas	15	70	16	80
SRC-NG-MEL	Site Reference Conditions, natural gas, GT at minimum emissions-compliant load (30 %, low-load extension)	gas	15	70	16	30
SUM-NG-100	Summer design (32.8 degC, 0.4 %), natural gas, base load	gas	32.8	45	24	100
WIN-NG-100	Winter design (-4.8 degC, 99.6 %), natural gas, base load (GT at output limit)	gas	-4.8	80	8	100

3. Results

Item	MAX-NG-100	MIN-NG-100	SRC-LDO-100	SRC-NG-060	SRC-NG-100	SRC-NG-GT80	SRC-NG-MEL	SUM-NG-100	WIN-NG-100
GT output MW	348.2	470.0	400.5	237.1	425.0	340.0	127.5	372.8	470.0
ST output MW	175.8	204.9	178.4	140.7	200.7	175.1	94.4	184.5	207.9
Gross output MW	524.1	674.9	578.9	377.8	625.7	515.0	221.9	557.3	677.9
Auxiliaries + transformer losses MW	10.8	12.1	11.1	9.3	11.8	10.5	8.1	11.1	12.2
Net output MW	513.3	662.8	567.8	368.5	613.9	504.5	213.8	546.2	665.7
Heat input LHV MW	832.2	1,058.3	961.2	635.1	983.0	827.3	412.9	881.2	1,063.8
of which other plant fuel consumers MW	0.8	2.3	0.0	1.0	1.5	1.3	0.6	0.8	2.3
Net heat rate LHV kJ/kWh	5,836.4	5,748.2	6,094.4	6,204.0	5,764.2	5,903.5	6,952.8	5,807.7	5,752.8
Net efficiency %	61.7	62.6	59.1	58.0	62.5	61.0	51.8	62.0	62.6

4. Checks against guarantees and limits

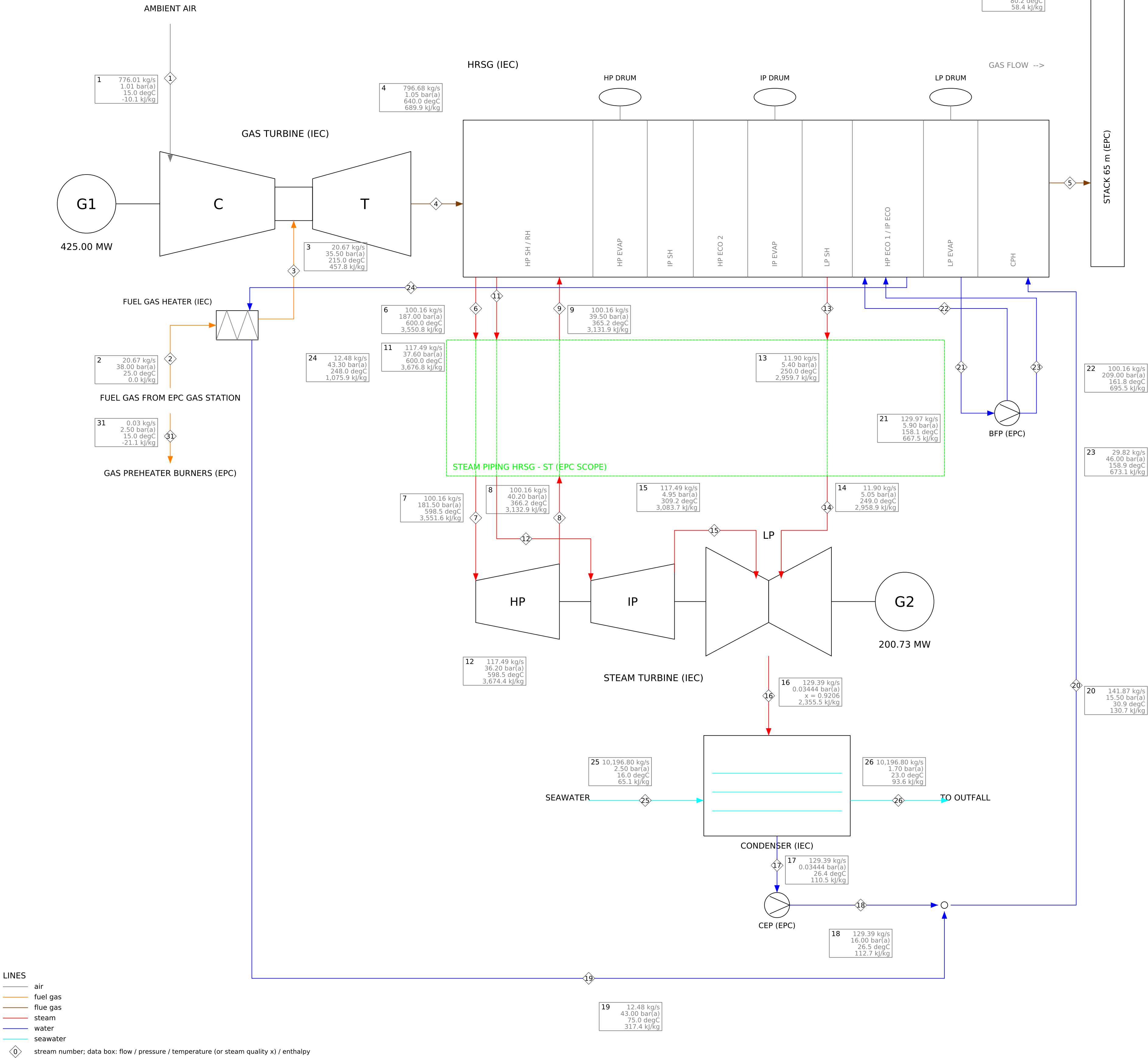
Case	Result	Check
MIN-NG-100	INFO	net export 662.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)
SRC-LDO-100	OK	guarantee cover: at IG-09/IG-10 limits and this HMB's auxiliaries the net plant is 560.9 MW / 6,139 kJ/kWh vs Owner PG-15 560 MW / PG-16 6,150 kJ/kWh
SRC-NG-100	OK	guarantee cover: at IG-01/IG-02 limits and this HMB's auxiliaries the net plant is 603.2 MW / 5,847 kJ/kWh vs Owner PG-01 600 MW / PG-02 5,860 kJ/kWh
WIN-NG-100	INFO	net export 665.7 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

5. Auxiliary loads at Site Reference Conditions

Id	Consumer	Basis	kW
AL-01	GT package auxiliaries (lube oil, hydraulic, enclosure ventilation, fuel skid, filter house)	fixed	1,180
AL-02	GT generator auxiliaries (seal oil, stator water, excitation)	fixed	420
AL-03	ST auxiliaries (lube oil, EHC, gland steam exhausters)	fixed	380
AL-04	ST generator auxiliaries (excitation, TEWAC)	fixed	260
AL-05	HRSG auxiliaries (CPH recirculation pump, ammonia skid, drains)	fixed	210
AL-06	Condenser vacuum pump (1 running)	fixed	75
AL-07	Turbine control system and UPS load	fixed	60
AL-08	GT GSU losses at this case load	fixed	1,107
AL-09	ST GSU losses at this case load	fixed	584
AL-10	Boiler feed pumps 2 x 100 % (1 running)	enthalpy rise 2,981 kW / motor 0.965	3,089
AL-11	Condensate extraction pumps 2 x 100 % (1 running)	enthalpy rise 282 kW / motor 0.95	297
AL-12	Circulating water pumps 2 x 50 %	10,197 kg/s x g x 16.0 m / (0.86 x 0.96)	1,939
AL-13	Closed cooling water pumps	fixed	160
AL-14	Electro-chlorination	fixed	180
AL-15	Water treatment (SWRO, demineralisation, condensate polishing)	fixed	250
AL-16	Compressed air	fixed	180
AL-17	HVAC and lighting	fixed	750
AL-18	Ammonia and chemical dosing	fixed	40
AL-19	Fuel gas receiving station	fixed	60
AL-20	LDO system (standby on gas)	fixed	30
AL-21	UAT and MV/LV transformer losses	fixed	240
AL-22	Miscellaneous (workshop, lighting outdoor, cathodic protection)	fixed	300

6. Notes

- Performance cases are at continuous blowdown closed (as during the performance tests); the demineralised water make-up, blowdown and cycle losses in normal operation are given in the Plant Water Balance ALP-EPC-00000-PR-STU-0001.
- The station gas preheaters (indirect water-bath heaters, DEC-EPCE-0014) burn plant fuel downstream of the fiscal metering: their fuel (stream 31) is part of the plant heat input and of the net heat rate, not of the power island (gross) heat rate.
- Auxiliary loads of the balance of plant follow the preliminary equipment sizing (DEC-EPCE-0014 / 0015 / 0016) and are updated with the electrical load list.
- The diagrams of all cases follow this report (A1 sheets, stream data in the boxes, node balances in the case workbooks ALP-HMB-<case>.xlsx).



STREAM TABLE - SRC-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	776.01	1.01	15.0	-10.1	-7.8
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	20.67	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	20.67	35.50	215.0	457.8	9.5
4	GT exhaust gas to HRSG inlet	796.68	1.05	640.0	689.9	549.6
5	Flue gas at HRSG outlet to stack	796.68	1.01	80.2	58.4	46.5
6	HP steam at HRSG outlet (HP SH outlet)	100.16	187.00	600.0	3,550.8	355.6
7	HP steam at ST main stop valve inlet	100.16	181.50	598.5	3,551.6	355.7
8	Cold reheat steam at HP turbine exhaust	100.16	40.20	366.2	3,132.9	313.8
9	Cold reheat steam at HRSG reheater inlet	100.16	39.50	365.2	3,131.9	313.7
10	IP superheated steam (joins cold reheat inside the HRSG)	17.34	40.30	330.0	3,041.7	52.7
11	Hot reheat steam at HRSG outlet	117.49	37.60	600.0	3,676.8	432.0
12	Hot reheat steam at ST IP stop/control valve inlet	117.49	36.20	598.5	3,674.4	431.7
13	LP steam at HRSG outlet	11.90	5.40	250.0	2,959.7	35.2
14	LP steam at ST LP admission	11.90	5.05	249.0	2,958.9	35.2
15	IP turbine exhaust (crossover, internal)	117.49	4.95	309.2	3,083.7	362.3
16	LP turbine exhaust to condenser (UEEP)	129.39	0.0344	x 0.921	2,355.5	304.8
17	Condensate at hotwell outlet / CEP suction	129.39	0.0344	26.4	110.5	14.3
18	Condensate at CEP discharge	129.39	16.00	26.5	112.7	14.6
19	Fuel gas heater water return to condensate	12.48	43.00	75.0	317.4	4.0
20	Condensate to HRSG condensate preheater inlet	141.87	15.50	30.9	130.7	18.5
21	Feedwater from LP drum to boiler feed pumps	129.97	5.90	158.1	667.5	86.8
22	HP feedwater at BFP discharge to HP economiser	100.16	209.00	161.8	695.5	69.7
23	IP feedwater at BFP interstage to IP economiser	29.82	46.00	158.9	673.1	20.1
24	IP economiser water to performance fuel gas heater	12.48	43.30	248.0	1,075.9	13.4
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	23.0	93.6	954.4
31	Fuel gas to the station preheater burners (water-bath heaters)	0.03	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

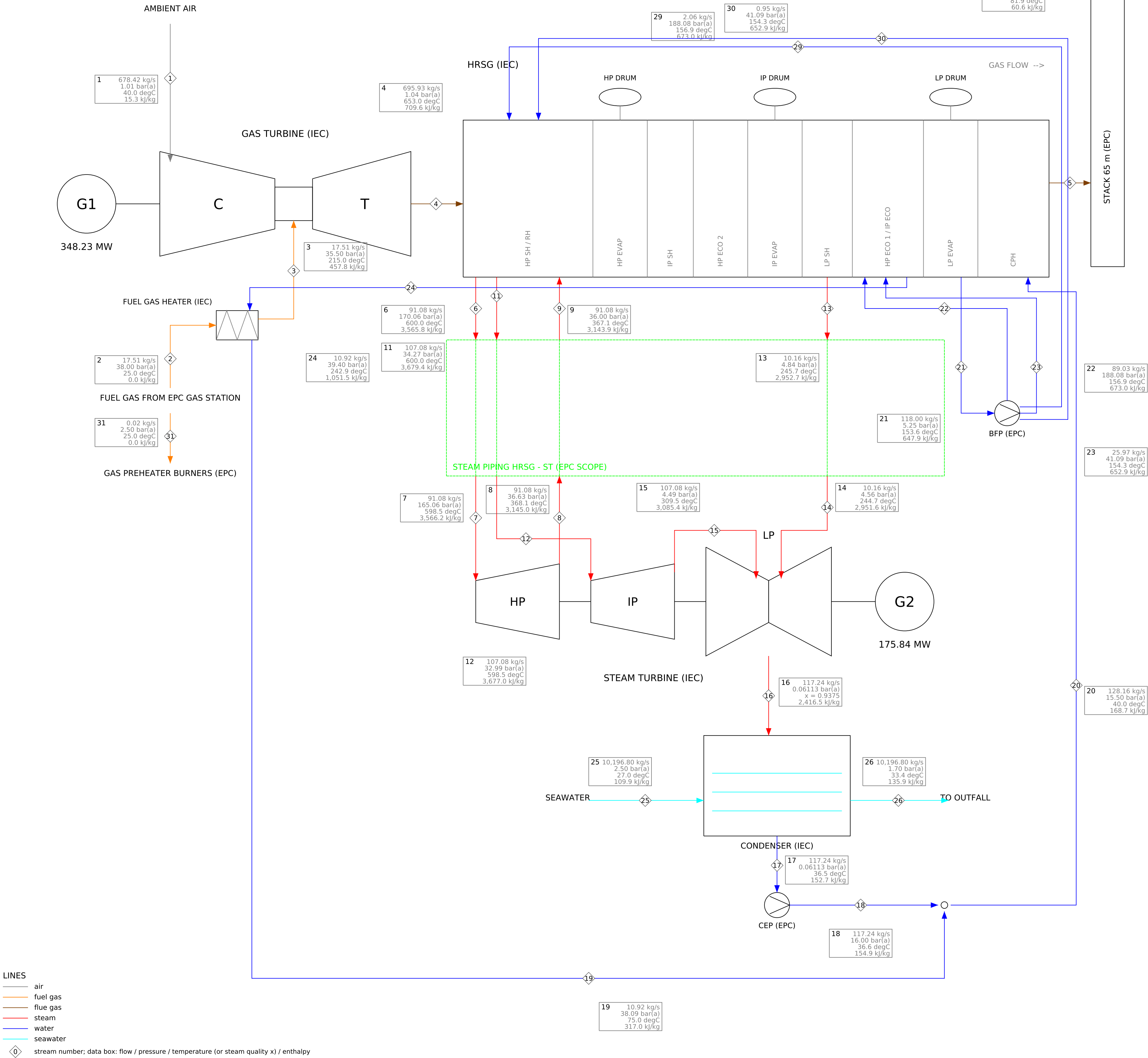
PLANT POWER BALANCE

GT output (generator terminals)	425.00	MW
ST output (generator terminals)	200.73	MW
Gross output	625.73	MW
Auxiliary loads	-9.86	MW
Transformer losses (GSU, UAT, MV/LV)	-1.93	MW
NET OUTPUT (HV terminals)	613.94	MW
Heat input (LHV)	983.02	MW
Gross heat rate (LHV)	5,647	kJ/kWh
NET HEAT RATE (LHV)	5,764	kJ/kWh
Net efficiency (LHV)	62.45	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	101,752	Sm3/h
Circulating water flow	36,112	m3/h
GT load	100.00	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.004 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.027 MW (tolerance 0.97 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.018 MW (tolerance 0.98 MW)
OK	energy balance HRSG: residual +2.039 MW vs stated casing loss 2.012 MW (tolerance 0.97 MW)
OK	energy balance PIPE: residual +0.298 MW (tolerance 1.14 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.82 MW)
OK	fuel LHV from the composition (ISO 6976) 47.48 MJ/kg vs DP-CASE-0018 47.48 MJ/kg (-0.00 %)
OK	auxiliary + transformer losses 1.88 % of gross vs DP-CASE-0031 <= 2.2 %
OK	condenser pressure 34.44 mbar(a) vs DP-OWNR-0021 <= 36.0 mbar(a)
OK	gas flow 101,752 Sm3/h = 84.8 % of DP-OWNR-0033 (120,000 Sm3/h)
OK	net export 613.9 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)
OK	gross output 625.73 MW vs IG-01 >= 615.00 (margin +1.74 %)
OK	gross heat rate 5,646.88 kJ/kWh vs IG-02 <= 5,735.00 (margin +1.54 %)
OK	GT output 425.00 MW vs IG-03 >= 418.00 (margin +1.67 %)
OK	condenser pressure 34.44 mbar(a) vs IG-04 <= 35.50 (margin +2.99 %)
OK	net output 613.94 MW vs PG-01 >= 600.00 (margin +2.32 %)
OK	net heat rate 5,764.23 kJ/kWh vs PG-02 <= 5,860.00 (margin +1.63 %)
OK	CW temperature rise 7.00 K vs PG-13 <= 7.00 (margin +0.00 %)
OK	guarantee cover: at IG-01/IG-02 limits and this HMB's auxiliaries the net plant is 603.2 MW / 5,847 kJ/kWh vs Owner PG-01 600 MW / PG-02 5,860 kJ/kWh

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-100
CASE	Site Reference Conditions, natural gas, base load (guarantee case)
DOCUMENT	ALP-HMB-SRC-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - MAX-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	678.42	1.01	40.0	15.3	10.4
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	17.51	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	17.51	35.50	215.0	457.8	8.0
4	GT exhaust gas to HRSG inlet	695.93	1.04	653.0	709.6	493.8
5	Flue gas at HRSG outlet to stack	695.93	1.01	81.9	60.6	42.2
6	HP steam at HRSG outlet (HP SH outlet)	91.08	170.06	600.0	3,565.8	324.8
7	HP steam at ST main stop valve inlet	91.08	165.06	598.5	3,566.2	324.8
8	Cold reheat steam at HP turbine exhaust	91.08	36.63	368.1	3,145.0	286.5
9	Cold reheat steam at HRSG reheater inlet	91.08	36.00	367.1	3,143.9	286.4
10	IP superheated steam (joins cold reheat inside the HRSG)	15.05	36.66	327.0	3,043.8	45.8
11	Hot reheat steam at HRSG outlet	107.08	34.27	600.0	3,679.4	394.0
12	Hot reheat steam at ST IP stop/control valve inlet	107.08	32.99	598.5	3,677.0	393.7
13	LP steam at HRSG outlet	10.16	4.84	245.7	2,952.7	30.0
14	LP steam at ST LP admission	10.16	4.56	244.7	2,951.6	30.0
15	IP turbine exhaust (crossover, internal)	107.08	4.49	309.5	3,085.4	330.4
16	LP turbine exhaust to condenser (UEEP)	117.24	0.0611	x 0.938	2,416.5	283.3
17	Condensate at hotwell outlet / CEP suction	117.24	0.0611	36.5	152.7	17.9
18	Condensate at CEP discharge	117.24	16.00	36.6	154.9	18.2
19	Fuel gas heater water return to condensate	10.92	38.09	75.0	317.0	3.5
20	Condensate to HRSG condensate preheater inlet	128.16	15.50	40.0	168.7	21.6
21	Feedwater from LP drum to boiler feed pumps	118.00	5.25	153.6	647.9	76.4
22	HP feedwater at BFP discharge to HP economiser	89.03	188.08	156.9	673.0	59.9
23	IP feedwater at BFP interstage to IP economiser	25.97	41.09	154.3	652.9	17.0
24	IP economiser water to performance fuel gas heater	10.92	39.40	242.9	1,051.5	11.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	27.0	109.9	1,120.4
26	Circulating seawater from condenser outlet	10,196.80	1.70	33.4	135.9	1,385.9
29	HP attemper spray water (BFP discharge)	2.06	188.08	156.9	673.0	1.4
30	Reheat attemper spray water (BFP interstage)	0.95	41.09	154.3	652.9	0.6
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	25.0	0.0	0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

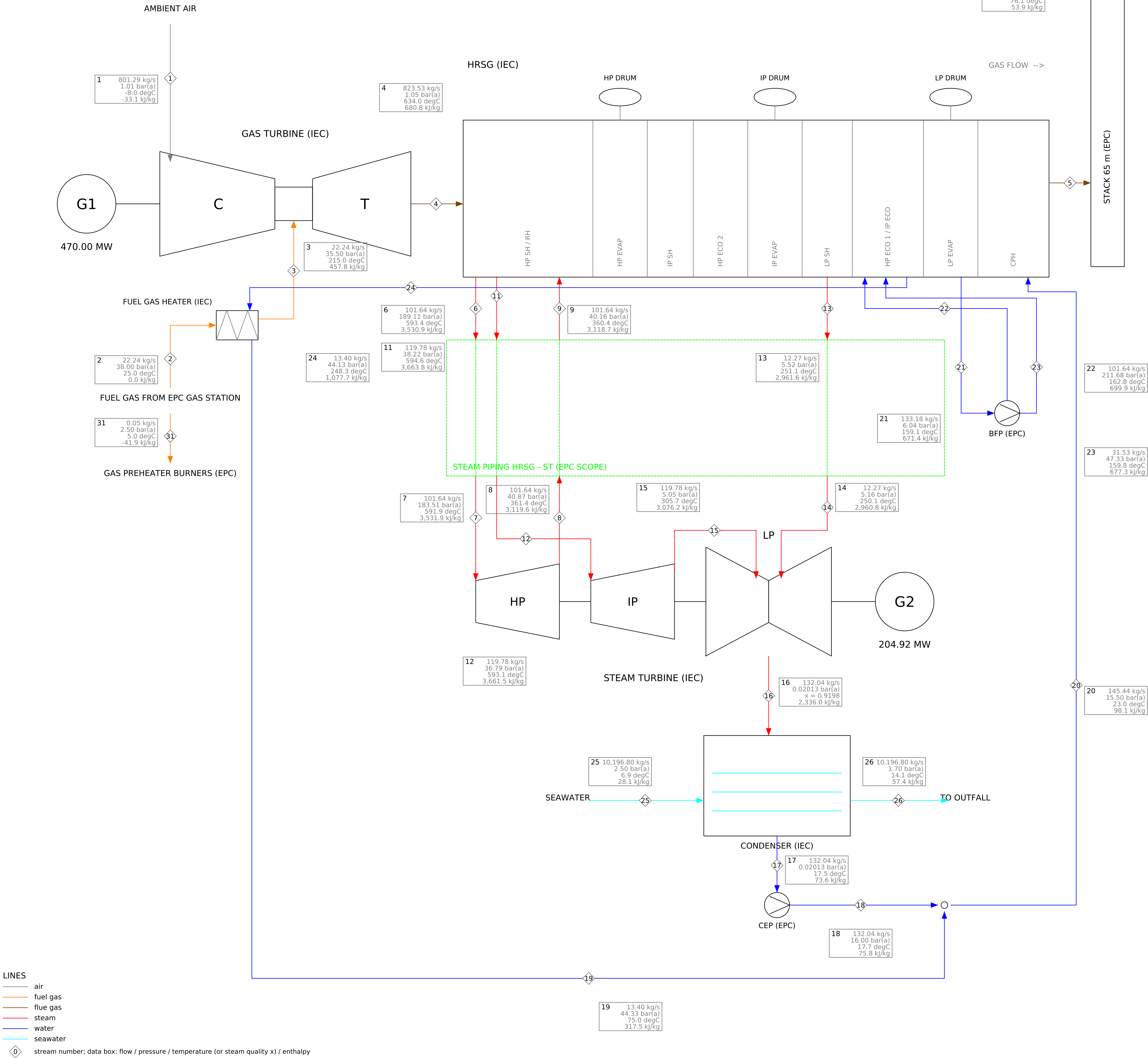
PLANT POWER BALANCE

GT output (generator terminals)	348.23	MW
ST output (generator terminals)	175.84	MW
Gross output	524.07	MW
Auxiliary loads	-9.26	MW
Transformer losses (GSU, UAT, MV/LV)	-1.52	MW
NET OUTPUT (HV terminals)	513.29	MW
Heat input (LHV)	832.17	MW
Gross heat rate (LHV)	5,711	kJ/kWh
NET HEAT RATE (LHV)	5,836	kJ/kWh
Net efficiency (LHV)	61.68	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	86,136	Sm ³ /h
Circulating water flow	36,190	m ³ /h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	83.61	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.164 MW (tolerance 1.40 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.005 MW (tolerance 0.85 MW)
OK	energy balance HRSG: residual +1.830 MW vs stated casing loss 1.807 MW (tolerance 0.88 MW)
OK	energy balance PIPE: residual +0.331 MW (tolerance 1.04 MW)
OK	energy balance ST: residual -0.000 MW (tolerance 0.75 MW)
OK	gas flow 86,136 Sm ³ /h = 71.8 % of DP-QWNR-0033 (120,000 Sm ³ /h)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - MAX-NG-100
CASE	Maximum ambient (40 degC) with maximum seawater (27 degC), natural gas, base loa
DOCUMENT	ALP-HMB-MAX-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - MIN-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	801.29	1.01	-8.0	-33.1	-26.5
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	22.24	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	22.24	35.50	215.0	457.8	10.2
4	GT exhaust gas to HRSG inlet	823.53	1.05	634.0	680.8	560.7
5	Flue gas at HRSG outlet to stack	823.53	1.01	76.1	53.9	44.4
6	HP steam at HRSG outlet (HP SH outlet)	101.64	189.11	593.4	3,530.9	358.9
7	HP steam at ST main stop valve inlet	101.64	183.51	591.9	3,531.9	359.0
8	Cold reheat steam at HP turbine exhaust	101.64	40.87	361.4	3,119.6	317.1
9	Cold reheat steam at HRSG reheater inlet	101.64	40.16	360.4	3,118.7	317.0
10	IP superheated steam (joins cold reheat inside the HRSG)	18.13	41.02	329.4	3,038.2	55.1
11	Hot reheat steam at HRSG outlet	119.78	38.22	594.6	3,663.8	438.8
12	Hot reheat steam at ST IP stop/control valve inlet	119.78	36.79	593.1	3,661.5	438.6
13	LP steam at HRSG outlet	12.27	5.52	251.1	2,961.6	36.3
14	LP steam at ST LP admission	12.27	5.16	250.1	2,960.8	36.3
15	IP turbine exhaust (crossover, internal)	119.78	5.05	305.7	3,076.2	368.5
16	LP turbine exhaust to condenser (UEEP)	132.04	0.0201	x 0.920	2,336.0	308.5
17	Condensate at hotwell outlet / CEP suction	132.04	0.0201	17.5	73.6	9.7
18	Condensate at CEP discharge	132.04	16.00	17.7	75.8	10.0
19	Fuel gas heater water return to condensate	13.40	44.33	75.0	317.5	4.3
20	Condensate to HRSG condensate preheater inlet	145.44	15.50	23.0	98.1	14.3
21	Feedwater from LP drum to boiler feed pumps	133.18	6.04	159.1	671.4	89.4
22	HP feedwater at BFP discharge to HP economiser	101.64	211.68	162.8	699.9	71.1
23	IP feedwater at BFP interstage to IP economiser	31.53	47.33	159.8	677.3	21.4
24	IP economiser water to performance fuel gas heater	13.40	44.13	248.3	1,077.7	14.4
25	Circulating seawater to condenser inlet	10,196.80	2.50	6.9	28.1	286.3
26	Circulating seawater from condenser outlet	10,196.80	1.70	14.1	57.4	585.1
31	Fuel gas to the station preheater burners (water-bath heaters)	0.05	2.50	5.0	-41.9	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

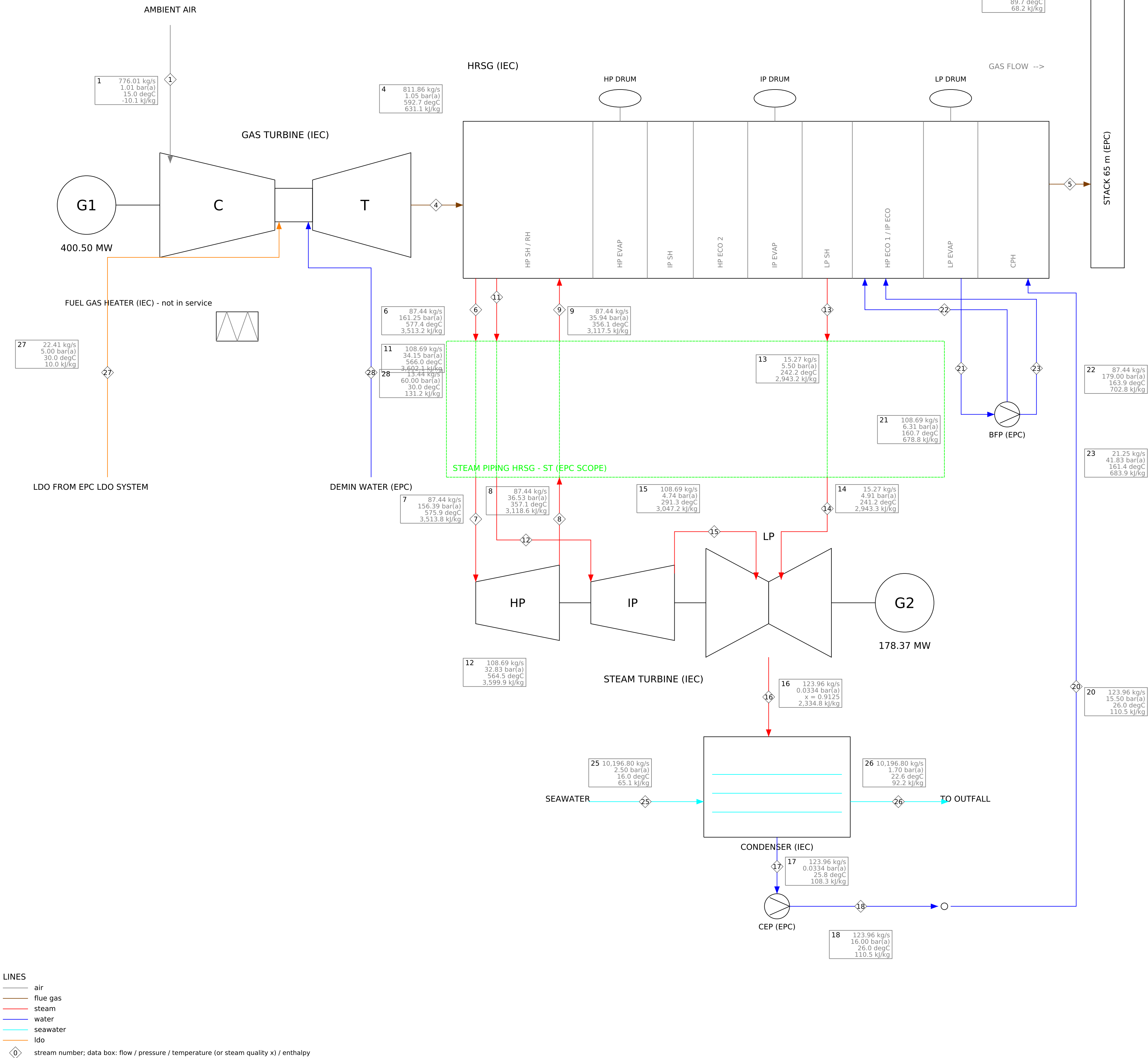
PLANT POWER BALANCE

GT output (generator terminals)	470.00	MW
ST output (generator terminals)	204.92	MW
Gross output	674.92	MW
Auxiliary loads	-9.97	MW
Transformer losses (GSU, UAT, MV/LV)	-2.15	MW
NET OUTPUT (HV terminals)	662.80	MW
Heat input (LHV)	1,058	MW
Gross heat rate (LHV)	5,633	kJ/kWh
NET HEAT RATE (LHV)	5,748	kJ/kWh
Net efficiency (LHV)	62.63	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	109,544	Sm3/h
Circulating water flow	36,047	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	107.96	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.032 MW (tolerance 0.59 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.007 MW (tolerance 1.04 MW)
OK	energy balance HRSG: residual +2.088 MW vs stated casing loss 2.065 MW (tolerance 0.98 MW)
OK	energy balance PIPE: residual +0.274 MW (tolerance 1.15 MW)
OK	energy balance ST: residual -0.003 MW (tolerance 0.83 MW)
OK	gas flow 109,544 Sm3/h = 91.3 % of DP-OWNR-0033 (120,000 Sm3/h)
INFO	net export 662.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - MIN-NG-100
CASE	Minimum ambient for full capability (-8 degC), natural gas, base load
DOCUMENT	ALP-HMB-MIN-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - SRC-LDO-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	776.01	1.01	15.0	-10.1	-7.8
4	GT exhaust gas to HRSG inlet	811.86	1.05	592.7	631.1	512.4
5	Flue gas at HRSG outlet to stack	811.86	1.01	89.7	68.2	55.4
6	HP steam at HRSG outlet (HP SH outlet)	87.44	161.25	577.4	3,513.2	307.2
7	HP steam at ST main stop valve inlet	87.44	156.39	575.9	3,513.8	307.2
8	Cold reheat steam at HP turbine exhaust	87.44	36.53	357.1	3,118.6	272.7
9	Cold reheat steam at HRSG reheater inlet	87.44	35.94	356.1	3,117.5	272.6
10	IP superheated steam (joins cold reheat inside the HRSG)	21.25	37.26	312.6	3,004.7	63.9
11	Hot reheat steam at HRSG outlet	108.69	34.15	566.0	3,602.1	391.5
12	Hot reheat steam at ST IP stop/control valve inlet	108.69	32.83	564.5	3,599.9	391.3
13	LP steam at HRSG outlet	15.27	5.50	242.2	2,943.2	44.9
14	LP steam at ST LP admission	15.27	4.91	241.2	2,943.3	44.9
15	IP turbine exhaust (crossover, internal)	108.69	4.74	291.3	3,047.2	331.2
16	LP turbine exhaust to condenser (UEEP)	123.96	0.0334	x 0.913	2,334.8	289.4
17	Condensate at hotwell outlet / CEP suction	123.96	0.0334	25.8	108.3	13.4
18	Condensate at CEP discharge	123.96	16.00	26.0	110.5	13.7
20	Condensate to HRSG condensate preheater inlet	123.96	15.50	26.0	110.5	13.7
21	Feedwater from LP drum to boiler feed pumps	108.69	6.31	160.7	678.8	73.8
22	HP feedwater at BFP discharge to HP economiser	87.44	179.00	163.9	702.8	61.5
23	IP feedwater at BFP interstage to IP economiser	21.25	41.83	161.4	683.9	14.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	22.6	92.2	939.9
27	LDO at liquid fuel skid inlet (IEC terminal point IF-02)	22.41	5.00	30.0	10.0	0.2
28	Demineralised water for NOx water injection (IF-03)	13.44	60.00	30.0	131.2	1.8

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

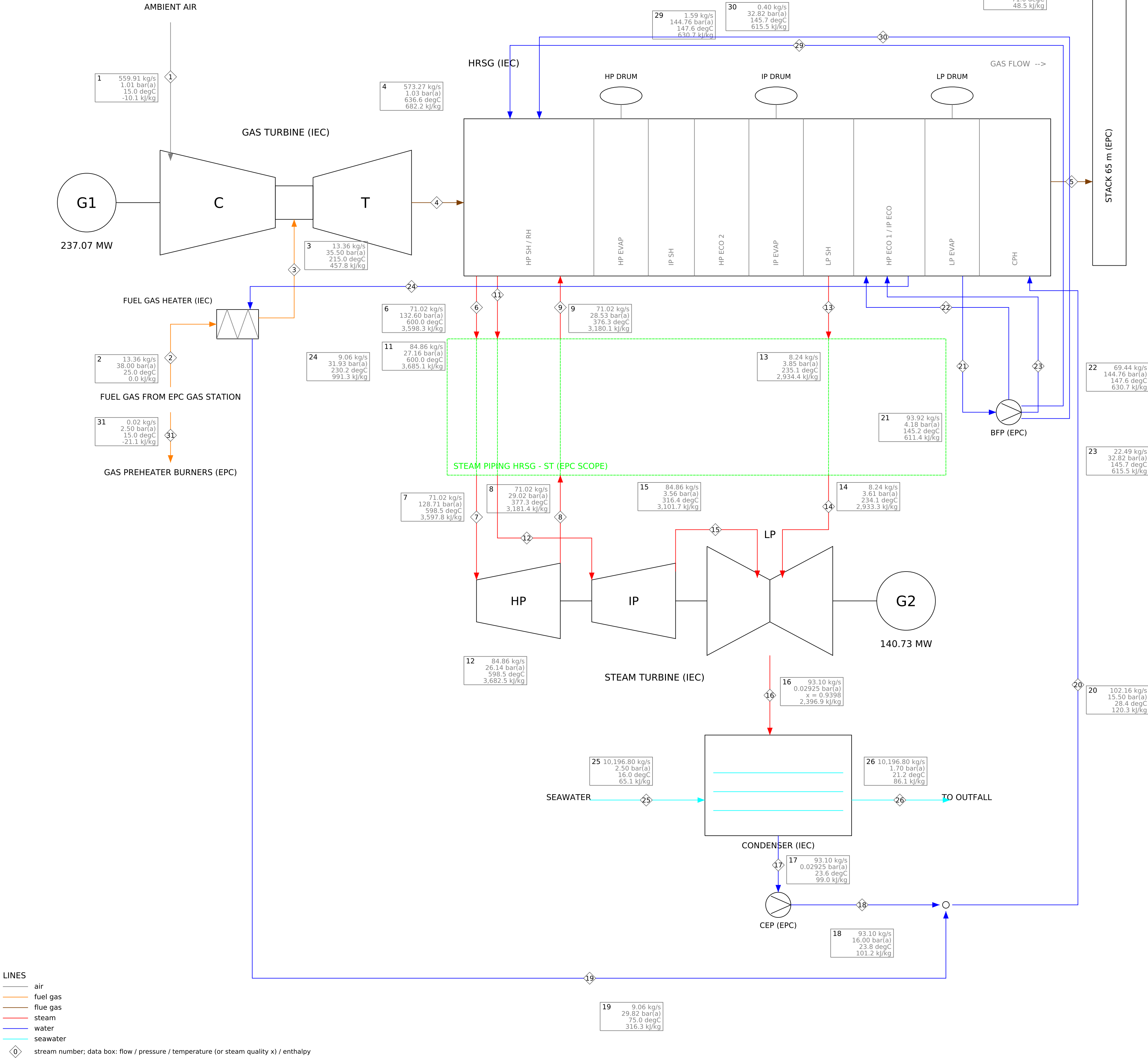
PLANT POWER BALANCE

GT output (generator terminals)	400.50	MW
ST output (generator terminals)	178.37	MW
Gross output	578.87	MW
Auxiliary loads	-9.35	MW
Transformer losses (GSU, UAT, MV/LV)	-1.73	MW
NET OUTPUT (HV terminals)	567.80	MW
Heat input (LHV)	961.22	MW
Gross heat rate (LHV)	5,978	kJ/kWh
NET HEAT RATE (LHV)	6,094	kJ/kWh
Net efficiency (LHV)	59.07	%
Fuel LHV (ISO 6976, from composition)	42.90	MJ/kg
Circulating water flow	36,112	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	92.48	%
Water injection (NOx)	13.44	kg/s

CHECKS (guarantees, limits, balances)

OK	energy balance CMX: residual +0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.047 MW (tolerance 0.95 MW)
OK	energy balance GT: residual +0.016 MW (tolerance 0.92 MW)
OK	energy balance HRSG: residual +1.860 MW vs stated casing loss 1.828 MW (tolerance 0.87 MW)
OK	energy balance PIPE: residual +0.286 MW (tolerance 1.02 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.74 MW)
OK	net export 567.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)
OK	gross output 578.87 MW vs IG-09 >= 572.00 (margin +1.20 %)
OK	gross heat rate 5,977.82 kJ/kWh vs IG-10 <= 6,020.00 (margin +0.70 %)
OK	net output 567.80 MW vs PG-15 >= 560.00 (margin +1.39 %)
OK	net heat rate 6,094.42 kJ/kWh vs PG-16 <= 6,150.00 (margin +0.90 %)
OK	guarantee cover: at IG-09/IG-10 limits and this HMB's auxiliaries the net plant is 560.9 MW / 6,139 kJ/kWh vs Owner PG-15 560 MW / PG-16 6,150 kJ/kWh

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-LDO-100
CASE	Site Reference Conditions, LDO with water injection, base load (guarantee PG-15/
DOCUMENT	ALP-HMB-SRC-LDO-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - SRC-NG-060

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	559.91	1.01	15.0	-10.1	-5.7
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	13.36	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	13.36	35.50	215.0	457.8	6.1
4	GT exhaust gas to HRSG inlet	573.27	1.03	636.6	682.2	391.1
5	Flue gas at HRSG outlet to stack	573.27	1.01	71.0	48.5	27.8
6	HP steam at HRSG outlet (HP SH outlet)	71.02	132.60	600.0	3,598.3	255.6
7	HP steam at ST main stop valve inlet	71.02	128.71	598.5	3,597.8	255.5
8	Cold reheat steam at HP turbine exhaust	71.02	29.02	377.3	3,181.4	226.0
9	Cold reheat steam at HRSG reheater inlet	71.02	28.53	376.3	3,180.1	225.9
10	IP superheated steam (joins cold reheat inside the HRSG)	13.43	29.19	310.9	3,024.1	40.6
11	Hot reheat steam at HRSG outlet	84.86	27.16	600.0	3,685.1	312.7
12	Hot reheat steam at ST IP stop/control valve inlet	84.86	26.14	598.5	3,682.5	312.5
13	LP steam at HRSG outlet	8.24	3.85	235.1	2,934.4	24.2
14	LP steam at ST LP admission	8.24	3.61	234.1	2,933.3	24.2
15	IP turbine exhaust (crossover, internal)	84.86	3.56	316.4	3,101.7	263.2
16	LP turbine exhaust to condenser (UEEP)	93.10	0.0293	x 0.940	2,396.9	223.1
17	Condensate at hotwell outlet / CEP suction	93.10	0.0293	23.6	99.0	9.2
18	Condensate at CEP discharge	93.10	16.00	23.8	101.2	9.4
19	Fuel gas heater water return to condensate	9.06	29.82	75.0	316.3	2.9
20	Condensate to HRSG condensate preheater inlet	102.16	15.50	28.4	120.3	12.3
21	Feedwater from LP drum to boiler feed pumps	93.92	4.18	145.2	611.4	57.4
22	HP feedwater at BFP discharge to HP economiser	69.44	144.76	147.6	630.7	43.8
23	IP feedwater at BFP interstage to IP economiser	22.49	32.82	145.7	615.5	13.8
24	IP economiser water to performance fuel gas heater	9.06	31.93	230.2	991.3	9.0
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	21.2	86.1	878.0
29	HP attemper spray water (BFP discharge)	1.59	144.76	147.6	630.7	1.0
30	Reheat attemper spray water (BFP interstage)	0.40	32.82	145.7	615.5	0.2
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

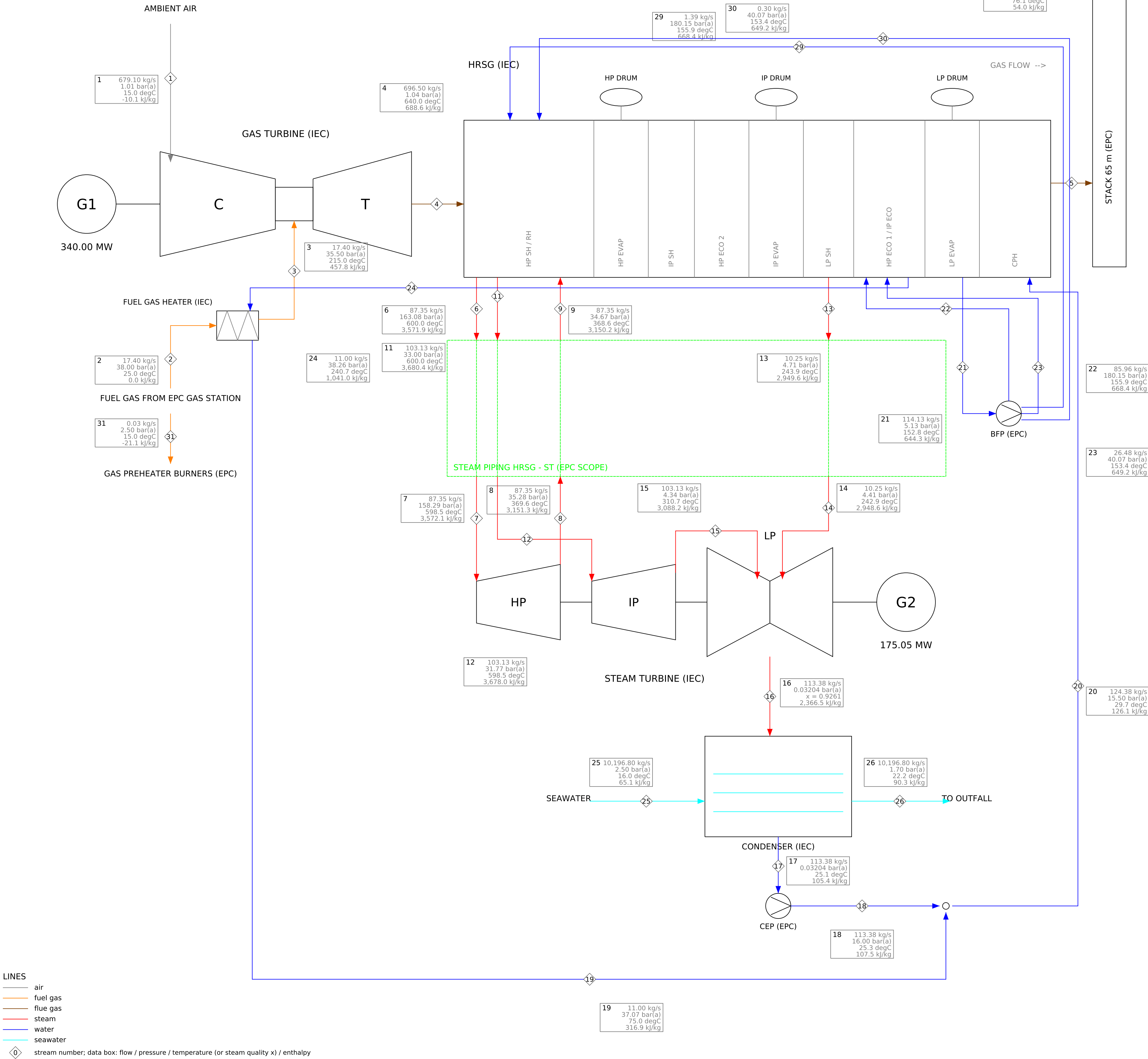
PLANT POWER BALANCE

GT output (generator terminals)	237.07	MW
ST output (generator terminals)	140.73	MW
Gross output	377.80	MW
Auxiliary loads	-8.20	MW
Transformer losses (GSU, UAT, MV/LV)	-1.06	MW
NET OUTPUT (HV terminals)	368.54	MW
Heat input (LHV)	635.13	MW
Gross heat rate (LHV)	6,043	kJ/kWh
NET HEAT RATE (LHV)	6,204	kJ/kWh
Net efficiency (LHV)	58.03	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	65,741	Sm ³ /h
Circulating water flow	36,112	m ³ /h
GT load	55.78	%
PLANT LOAD (NET, % OF SRC-NG-100)	60.03	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.004 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.184 MW (tolerance 0.89 MW)
OK	energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.013 MW (tolerance 0.63 MW)
OK	energy balance HRSG: residual +1.482 MW vs stated casing loss 1.453 MW (tolerance 0.69 MW)
OK	energy balance PIPE: residual +0.362 MW (tolerance 0.82 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.59 MW)
OK	gross heat rate 6,042.65 kJ/kWh vs IG-07 <= 6,060.00 (margin +0.29 %)
OK	net heat rate 6,204.01 kJ/kWh vs PG-03 <= 6,250.00 (margin +0.74 %)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-060
CASE	Site Reference Conditions, natural gas, 60 % plant net load (guarantee PG-03)
DOCUMENT	ALP-HMB-SRC-NG-060
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



LINES
— air
— fuel gas
— flue gas
— steam
— water
— seawater
◇ stream number; data box: flow / pressure / temperature (or steam quality x) / enthalpy

STREAM TABLE - SRC-NG-GT80

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	679.10	1.01	15.0	-10.1	-6.9
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	17.40	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	17.40	35.50	215.0	457.8	8.0
4	GT exhaust gas to HRSG inlet	696.50	1.04	640.0	688.6	479.6
5	Flue gas at HRSG outlet to stack	696.50	1.01	76.1	54.0	37.6
6	HP steam at HRSG outlet (HP SH outlet)	87.35	163.08	600.0	3,571.9	312.0
7	HP steam at ST main stop valve inlet	87.35	158.29	598.5	3,572.1	312.0
8	Cold reheat steam at HP turbine exhaust	87.35	35.28	369.6	3,151.3	275.3
9	Cold reheat steam at HRSG reheater inlet	87.35	34.67	368.6	3,150.2	275.2
10	IP superheated steam (joins cold reheat inside the HRSG)	15.47	35.40	323.1	3,037.3	47.0
11	Hot reheat steam at HRSG outlet	103.13	33.00	600.0	3,680.4	379.5
12	Hot reheat steam at ST IP stop/control valve inlet	103.13	31.77	598.5	3,678.0	379.3
13	LP steam at HRSG outlet	10.25	4.71	243.9	2,948.6	30.2
14	LP steam at ST LP admission	10.25	4.41	242.9	2,948.6	30.2
15	IP turbine exhaust (crossover, internal)	103.13	4.34	310.7	3,088.2	318.5
16	LP turbine exhaust to condenser (UEEP)	113.38	0.0320	x 0.926	2,366.5	268.3
17	Condensate at hotwell outlet / CEP suction	113.38	0.0320	25.1	105.4	11.9
18	Condensate at CEP discharge	113.38	16.00	25.3	107.5	12.2
19	Fuel gas heater water return to condensate	11.00	37.07	75.0	316.9	3.5
20	Condensate to HRSG condensate preheater inlet	124.38	15.50	29.7	126.1	15.7
21	Feedwater from LP drum to boiler feed pumps	114.13	5.13	152.8	644.3	73.5
22	HP feedwater at BFP discharge to HP economiser	85.96	180.15	155.9	668.4	57.5
23	IP feedwater at BFP interstage to IP economiser	26.48	40.07	153.4	649.2	17.2
24	IP economiser water to performance fuel gas heater	11.00	38.26	240.7	1,041.0	11.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	22.2	90.3	920.4
29	HP attenuation spray water (BFP discharge)	1.39	180.15	155.9	668.4	0.9
30	Reheat attenuation spray water (BFP interstage)	0.30	40.07	153.4	649.2	0.2
31	Fuel gas to the station preheater burners (water-bath heaters)	0.03	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

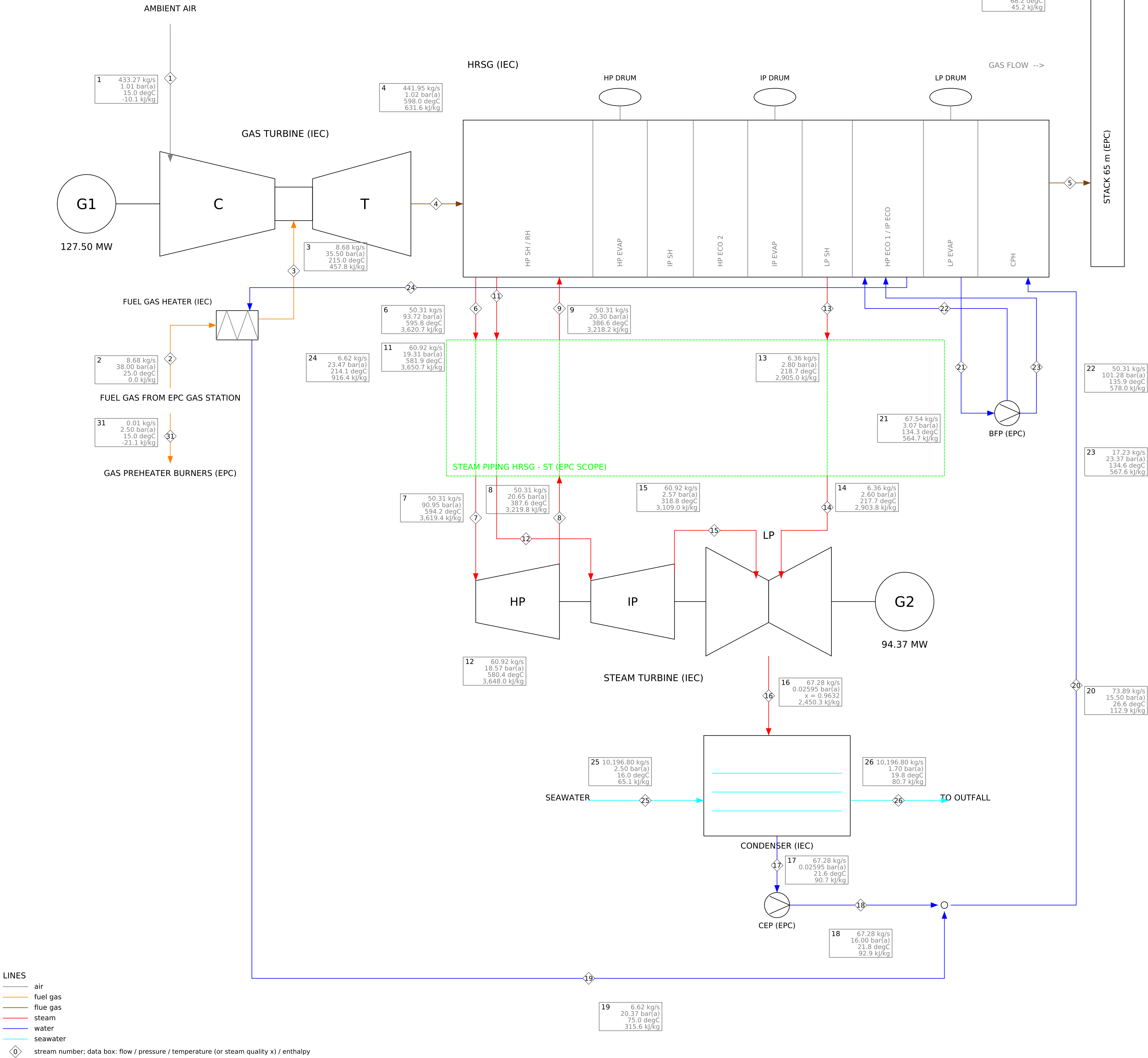
PLANT POWER BALANCE

GT output (generator terminals)	340.00	MW
ST output (generator terminals)	175.05	MW
Gross output	515.05	MW
Auxiliary loads	-9.05	MW
Transformer losses (GSU, UAT, MV/LV)	-1.49	MW
NET OUTPUT (HV terminals)	504.51	MW
Heat input (LHV)	827.33	MW
Gross heat rate (LHV)	5,774	kJ/kWh
NET HEAT RATE (LHV)	5,903	kJ/kWh
Net efficiency (LHV)	60.98	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	85,636	Sm ³ /h
Circulating water flow	36,112	m ³ /h
GT load	80.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	82.18	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.081 MW (tolerance 0.93 MW)
OK	energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.005 MW (tolerance 0.83 MW)
OK	energy balance HRSG: residual +1.789 MW vs stated casing loss 1.768 MW (tolerance 0.85 MW)
OK	energy balance PIPE: residual +0.342 MW (tolerance 1.00 MW)
OK	energy balance ST: residual +0.003 MW (tolerance 0.72 MW)
OK	gas flow 85,636 Sm ³ /h = 71.4 % of DP-OWNR-0033 (120,000 Sm ³ /h)
OK	net export 504.5 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-GT80
CASE	Site Reference Conditions, natural gas, GT at 80 % load
DOCUMENT	ALP-HMB-SRC-NG-GT80
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - SRC-NG-MEL

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	433.27	1.01	15.0	-10.1	-4.4
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	8.68	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	8.68	35.50	215.0	457.8	4.0
4	GT exhaust gas to HRSG inlet	441.95	1.02	598.0	631.6	279.1
5	Flue gas at HRSG outlet to stack	441.95	1.01	68.2	45.2	20.0
6	HP steam at HRSG outlet (HP SH outlet)	50.31	93.72	595.8	3,620.7	182.2
7	HP steam at ST main stop valve inlet	50.31	90.95	594.2	3,619.4	182.1
8	Cold reheat steam at HP turbine exhaust	50.31	20.65	387.6	3,219.8	162.0
9	Cold reheat steam at HRSG reheater inlet	50.31	20.30	386.6	3,218.2	161.9
10	IP superheated steam (joins cold reheat inside the HRSG)	10.61	20.89	291.3	3,001.3	31.8
11	Hot reheat steam at HRSG outlet	60.92	19.31	581.9	3,650.7	222.4
12	Hot reheat steam at ST IP stop/control valve inlet	60.92	18.57	580.4	3,648.0	222.2
13	LP steam at HRSG outlet	6.36	2.80	218.7	2,905.0	18.5
14	LP steam at ST LP admission	6.36	2.60	217.7	2,903.8	18.5
15	IP turbine exhaust (crossover, internal)	60.92	2.57	318.8	3,109.0	189.4
16	LP turbine exhaust to condenser (UEEP)	67.28	0.0260	x 0.963	2,450.3	164.8
17	Condensate at hotwell outlet / CEP suction	67.28	0.0260	21.6	90.7	6.1
18	Condensate at CEP discharge	67.28	16.00	21.8	92.9	6.3
19	Fuel gas heater water return to condensate	6.62	20.37	75.0	315.6	2.1
20	Condensate to HRSG condensate preheater inlet	73.89	15.50	26.6	112.9	8.3
21	Feedwater from LP drum to boiler feed pumps	67.54	3.07	134.3	564.7	38.1
22	HP feedwater at BFP discharge to HP economiser	50.31	101.28	135.9	578.0	29.1
23	IP feedwater at BFP interstage to IP economiser	17.23	23.37	134.6	567.6	9.8
24	IP economiser water to performance fuel gas heater	6.62	23.47	214.1	916.4	6.1
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	19.8	80.7	822.8
31	Fuel gas to the station preheater burners (water-bath heaters)	0.01	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

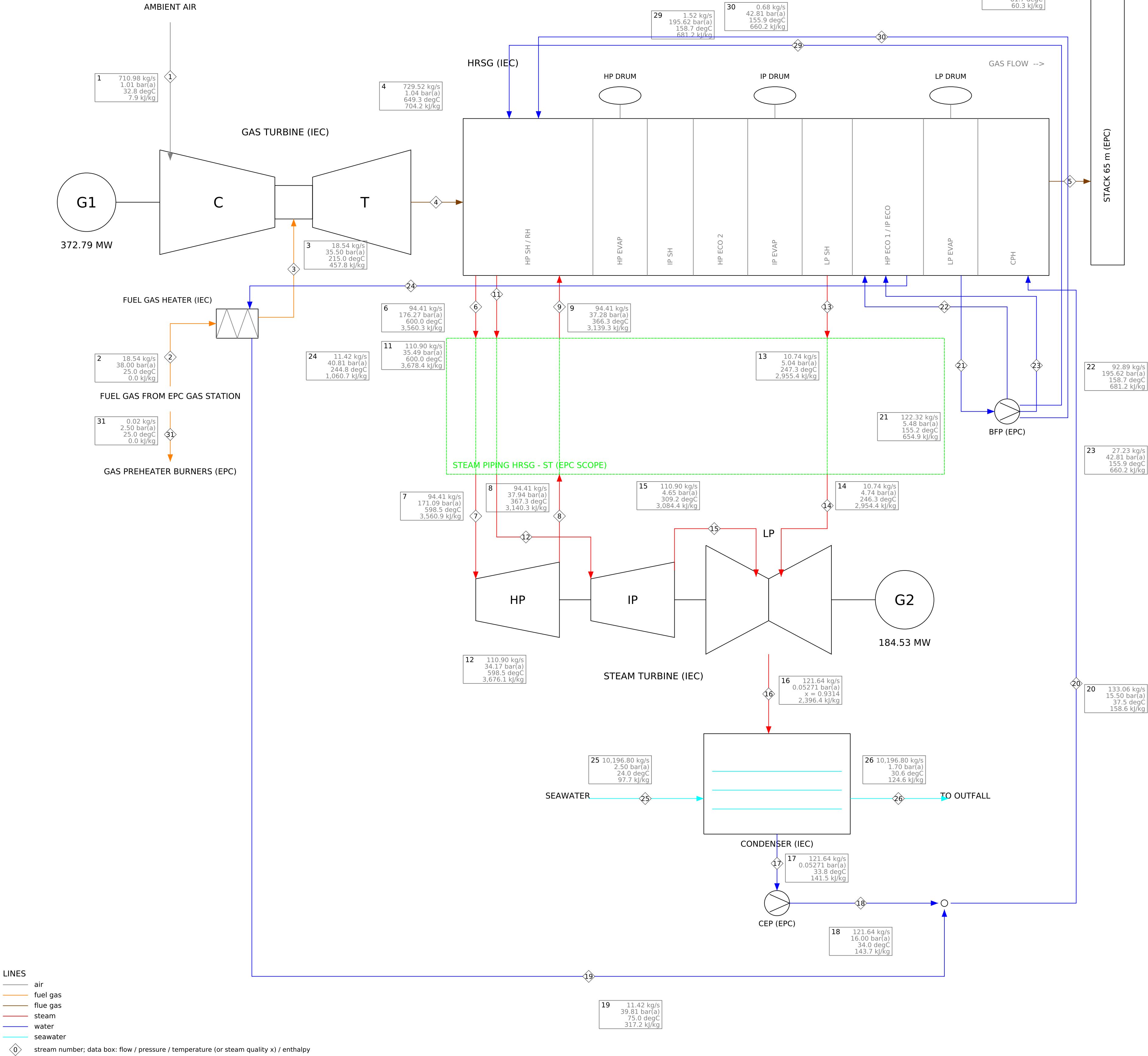
PLANT POWER BALANCE

GT output (generator terminals)	127.50	MW
ST output (generator terminals)	94.37	MW
Gross output	221.87	MW
Auxiliary loads	-7.38	MW
Transformer losses (GSU, UAT, MV/LV)	-0.73	MW
NET OUTPUT (HV terminals)	213.77	MW
Heat input (LHV)	412.86	MW
Gross heat rate (LHV)	6,689	kJ/kWh
NET HEAT RATE (LHV)	6,953	kJ/kWh
Net efficiency (LHV)	51.78	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	42,734	Sm3/h
Circulating water flow	36,112	m3/h
GT load	30.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	34.82	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.181 MW (tolerance 0.83 MW)
OK	energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.019 MW (tolerance 0.41 MW)
OK	energy balance HRSG: residual +1.048 MW vs stated casing loss 1.037 MW (tolerance 0.49 MW)
OK	energy balance PIPE: residual +0.322 MW (tolerance 0.59 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.42 MW)
OK	plant load (net, % of the reference case) 34.82 % of plant net output vs PG-11 <= 40.00 (margin +12.95 %)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-MEL
CASE	Site Reference Conditions, natural gas, GT at minimum emissions-compliant load (
DOCUMENT	ALP-HMB-SRC-NG-MEL
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



STREAM TABLE - SUM-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	710.98	1.01	32.8	7.9	5.6
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	18.54	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	18.54	35.50	215.0	457.8	8.5
4	GT exhaust gas to HRSG inlet	729.52	1.04	649.3	704.2	513.7
5	Flue gas at HRSG outlet to stack	729.52	1.01	81.7	60.3	44.0
6	HP steam at HRSG outlet (HP SH outlet)	94.41	176.27	600.0	3,560.3	336.1
7	HP steam at ST main stop valve inlet	94.41	171.09	598.5	3,560.9	336.2
8	Cold reheat steam at HP turbine exhaust	94.41	37.94	367.3	3,140.3	296.5
9	Cold reheat steam at HRSG reheater inlet	94.41	37.28	366.3	3,139.3	296.4
10	IP superheated steam (joins cold reheat inside the HRSG)	15.81	37.99	328.3	3,043.5	48.1
11	Hot reheat steam at HRSG outlet	110.90	35.49	600.0	3,678.4	407.9
12	Hot reheat steam at ST IP stop/control valve inlet	110.90	34.17	598.5	3,676.1	407.7
13	LP steam at HRSG outlet	10.74	5.04	247.3	2,954.4	31.7
14	LP steam at ST LP admission	10.74	4.74	246.3	2,954.4	31.7
15	IP turbine exhaust (crossover, internal)	110.90	4.65	309.2	3,084.4	342.1
16	LP turbine exhaust to condenser (UEEP)	121.64	0.0527	x 0.931	2,396.4	291.5
17	Condensate at hotwell outlet / CEP suction	121.64	0.0527	33.8	141.5	17.2
18	Condensate at CEP discharge	121.64	16.00	34.0	143.7	17.5
19	Fuel gas heater water return to condensate	11.42	39.81	75.0	317.2	3.6
20	Condensate to HRSG condensate preheater inlet	133.06	15.50	37.5	158.6	21.1
21	Feedwater from LP drum to boiler feed pumps	122.32	5.48	155.2	654.9	80.1
22	HP feedwater at BFP discharge to HP economiser	92.89	195.62	158.7	681.2	63.3
23	IP feedwater at BFP interstage to IP economiser	27.23	42.81	155.9	660.2	18.0
24	IP economiser water to performance fuel gas heater	11.42	40.81	244.8	1,060.7	12.1
25	Circulating seawater to condenser inlet	10,196.80	2.50	24.0	97.7	995.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	30.6	124.6	1,270.2
29	HP attemper spray water (BFP discharge)	1.52	195.62	158.7	681.2	1.0
30	Reheat attemper spray water (BFP interstage)	0.68	42.81	155.9	660.2	0.4
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	25.0	0.0	0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

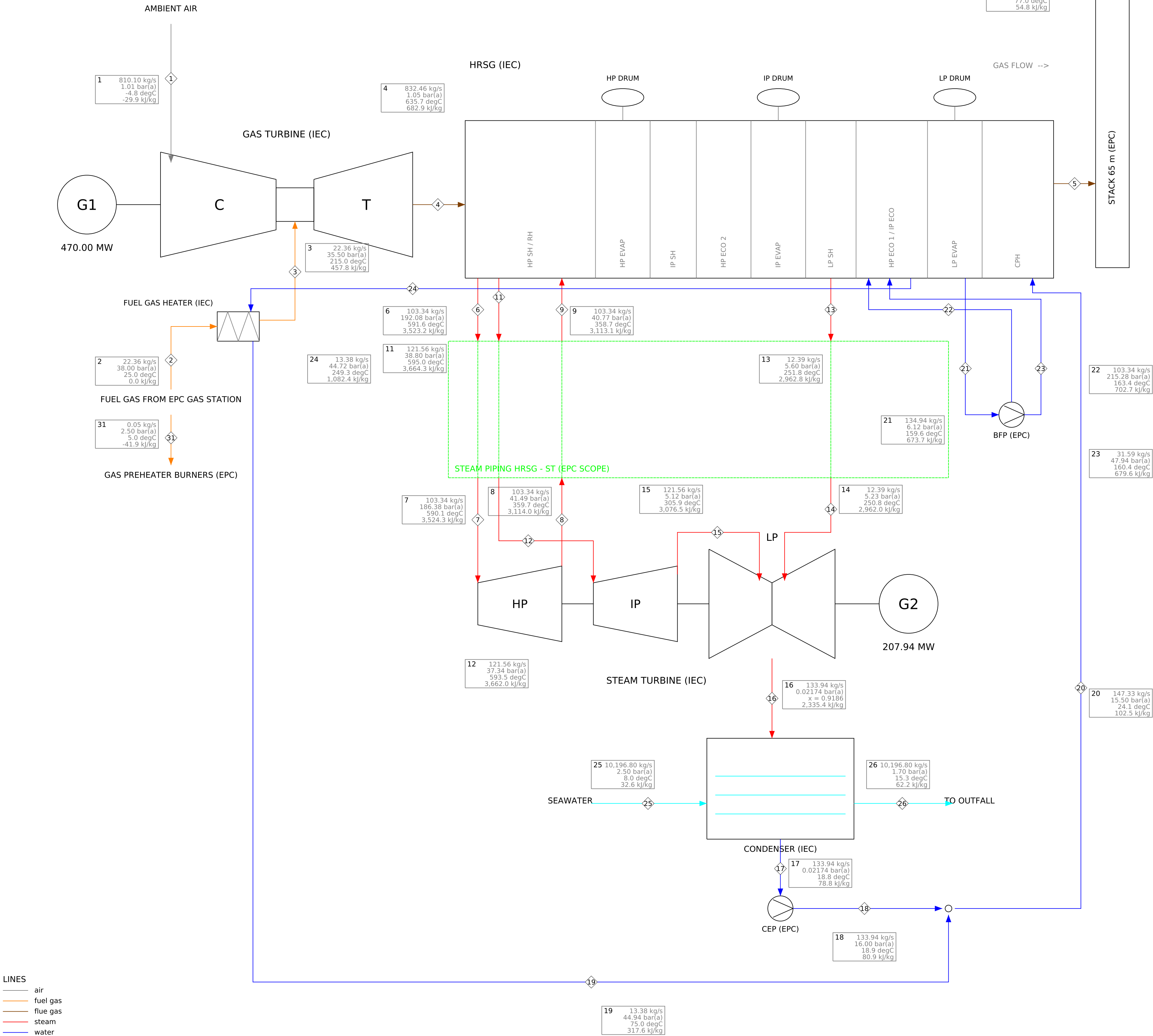
PLANT POWER BALANCE

GT output (generator terminals)	372.79	MW
ST output (generator terminals)	184.53	MW
Gross output	557.33	MW
Auxiliary loads	-9.47	MW
Transformer losses (GSU, UAT, MV/LV)	-1.65	MW
NET OUTPUT (HV terminals)	546.21	MW
Heat input (LHV)	881.16	MW
Gross heat rate (LHV)	5,686	kJ/kWh
NET HEAT RATE (LHV)	5,808	kJ/kWh
Net efficiency (LHV)	61.99	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	91,208	Sm ³ /h
Circulating water flow	36,169	m ³ /h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	88.97	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.002 MW (tolerance 1.29 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.001 MW (tolerance 0.89 MW)
OK	energy balance HRSG: residual +1.912 MW vs stated casing loss 1.879 MW (tolerance 0.91 MW)
OK	energy balance PIPE: residual +0.321 MW (tolerance 1.07 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.78 MW)
OK	gas flow 91,208 Sm ³ /h = 76.0 % of DP-QWNR-0033 (120,000 Sm ³ /h)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SUM-NG-100
CASE	Summer design (32.8 degC, 0.4 %), natural gas, base load
DOCUMENT	ALP-HMB-SUM-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit



LINES

- air
- fuel gas
- flue gas
- steam
- water
- seawater

◊ stream number; data box: flow / pressure / temperature (or steam quality x) / enthalpy

STREAM TABLE - WIN-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	810.10	1.01	-4.8	-29.9	-24.2
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	22.36	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	22.36	35.50	215.0	457.8	10.2
4	GT exhaust gas to HRSG inlet	832.46	1.05	635.7	682.9	568.5
5	Flue gas at HRSG outlet to stack	832.46	1.01	77.0	54.8	45.6
6	HP steam at HRSG outlet (HP SH outlet)	103.34	192.08	591.6	3,523.2	364.1
7	HP steam at ST main stop valve inlet	103.34	186.38	590.1	3,524.3	364.2
8	Cold reheat steam at HP turbine exhaust	103.34	41.49	359.7	3,114.0	321.8
9	Cold reheat steam at HRSG reheater inlet	103.34	40.77	358.7	3,113.1	321.7
10	IP superheated steam (joins cold reheat inside the HRSG)	18.21	41.62	330.4	3,039.2	55.3
11	Hot reheat steam at HRSG outlet	121.56	38.80	595.0	3,664.3	445.4
12	Hot reheat steam at ST IP stop/control valve inlet	121.56	37.34	593.5	3,662.0	445.1
13	LP steam at HRSG outlet	12.39	5.60	251.8	2,962.8	36.7
14	LP steam at ST LP admission	12.39	5.23	250.8	2,962.0	36.7
15	IP turbine exhaust (crossover, internal)	121.56	5.12	305.9	3,076.5	374.0
16	LP turbine exhaust to condenser (UEEP)	133.94	0.0217	x 0.919	2,335.4	312.8
17	Condensate at hotwell outlet / CEP suction	133.94	0.0217	18.8	78.8	10.6
18	Condensate at CEP discharge	133.94	16.00	18.9	80.9	10.8
19	Fuel gas heater water return to condensate	13.38	44.94	75.0	317.6	4.3
20	Condensate to HRSG condensate preheater inlet	147.33	15.50	24.1	102.5	15.1
21	Feedwater from LP drum to boiler feed pumps	134.94	6.12	159.6	673.7	90.9
22	HP feedwater at BFP discharge to HP economiser	103.34	215.28	163.4	702.7	72.6
23	IP feedwater at BFP interstage to IP economiser	31.59	47.94	160.4	679.6	21.5
24	IP economiser water to performance fuel gas heater	13.38	44.72	249.3	1,082.4	14.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	8.0	32.6	332.0
26	Circulating seawater from condenser outlet	10,196.80	1.70	15.3	62.2	634.0
31	Fuel gas to the station preheater burners (water-bath heaters)	0.05	2.50	5.0	-41.9	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	470.00	MW
ST output (generator terminals)	207.94	MW
Gross output	677.95	MW
Auxiliary loads	-10.07	MW
Transformer losses (GSU, UAT, MV/LV)	-2.17	MW
NET OUTPUT (HV terminals)	665.70	MW
Heat input (LHV)	1,064	MW
Gross heat rate (LHV)	5,637	kJ/kWh
NET HEAT RATE (LHV)	5,753	kJ/kWh
Net efficiency (LHV)	62.58	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	110,111	Sm3/h
Circulating water flow	36,055	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	108.43	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.003 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.183 MW (tolerance 0.64 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.020 MW (tolerance 1.05 MW)
OK	energy balance HRSG: residual +2.113 MW vs stated casing loss 2.091 MW (tolerance 1.00 MW)
OK	energy balance PIPE: residual +0.260 MW (tolerance 1.17 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.85 MW)
OK	gas flow 110,111 Sm3/h = 91.8 % of DP-OWNR-0033 (120,000 Sm3/h)
INFO	net export 665.7 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - WIN-NG-100
CASE	Winter design (-4.8 degC, 99.6 %), natural gas, base load (GT at output limit)
DOCUMENT	ALP-HMB-WIN-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit